Deep Learning

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Dataset :- It's a palmer Penguins from kaggle. It's a classification dataset with 3 species and exactly 4 ~~neural~~ numerical features, so it fits our 4-2-3 neural network.

Input layer → 4 neurons

$X = [39.1, 18.7, 181, 3750]$

X_1 = Bill/culmen length

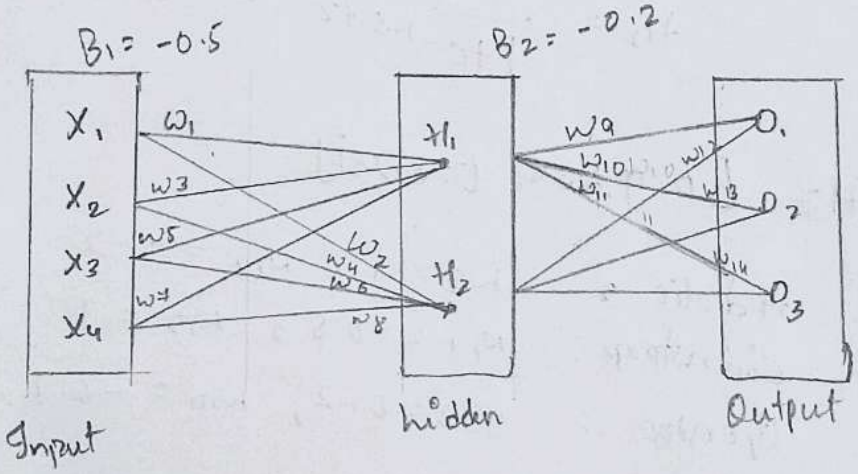
X_2 = Bill/culmen depth

X_3 = Flipper length

X_4 = Body Mass.

Output layer → 3 neurons.

Adelie, Chinstrap Gentoo



One Row :-

Input = [39.1, 18.7, 181, 3750]

Weights for H_1 = $W_1 = 0.05$, $W_3 = -0.08$, $W_5 = 0.01$, $W_7 = 0$

Weights for H_2 = $W_2 = -0.03$, $W_4 = 0.06$, $W_6 = 0.008$, $W_8 = 0$

Bias $B_1 = -0.5$

Bias $B_2 = -0.2$

Calculating H_1

$$Z_{H_1} = X_1 W_1 + X_2 W_3 + X_3 W_5 + X_4 W_7 + B_1$$
$$= (39.1 \times 0.05) + (18.4 \times -0.08) + (181 \times 0.01) + (3750 \times 0.0002) - 0.5$$
$$= 1.955 - 1.496 + 1.81 + 0.75 - 0.5$$

$$\boxed{Z_{H_1} = 2.519.}$$

Sigmoid for H_1

$$H_1 = \frac{1}{1 + e^{-2.519}} = \underline{0.9255}$$

Calculating H_2

$$Z_{H_2} = X_1 W_2 + X_2 W_4 + X_3 W_6 + X_4 W_8 + B_2$$
$$= (39.1 \times 0.03) + (18.7 \times 0.06) + (181 \times 0.008) + (3750 \times 0.0001) - 0.2$$
$$= -1.173 + 1.122 + 1.448 + 0.375 - 0.2$$

$$\boxed{Z_{H_2} = 1.572}$$

Sigmoid for H_2

$$H_2 = \frac{1}{1 + e^{-1.572}} = 0.8281.$$

$$\therefore H = [0.9255,\ 0.8281]$$

Output layers :- Adelie : $W_9 = 1$, $W_{10} = 0.2$

Chinstrap : $W_{11} = -0.5$, $W_{12} = 1$

Gentoo : $W_{13} = 0.2$, $W_{14} = -0.1$

for Adelie $Z = H_1 W_9 + H_2 W_{10}$
$$= (0.9255 \times 1) + (0.8281 \times -0.2)$$
$$= 0.9255 - 0.1656$$
$$= \underline{0.7599}$$

for Chinstrap $z = H_1 w_{11} + H_2 w_{12} - 0.1$

$= (0.9255 \times 0.5) + (0.8281 \times 1) - 0.1$

$= -0.4628 + 0.8281 - 0.1$

$= \underline{0.2653}$

for Gentoo $z = H_1 w_{13} + H_2 w_{14} - 0.05$

$= (0.9255 \times 0.2) + (0.8281 \times -0.1) - 0.05$

$= 0.1851 - 0.0828 - 0.05$

$= \underline{0.0523}$

Output = [0.7599, 0.2653, 0.0523]

Highest Score = 0.7599

∴ Predicted class = Adelie.

final answer = Adelie